

Course Code : ECT 456-3

MNQR/RS – 23 / 1095

**Eighth Semester B. E. (Electronics and Communication Engineering)
Examination**

ANTENNA THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) All questions carry marks as indicated.

1. (a) Elaborate in brief important parameters of antenna (at least 5 features).
5(CO1)
- (b) Calculate the maximum effective aperture of a microwave antenna which has directivity of 900. Also calculate efficiency of the antenna if its radiation resistance is $80\ \Omega$ and load resistance is $20\ \Omega$.
5(CO1)
2. (a) Explain the field equivalence principle for antenna with suitable example.
5(CO4)
- (b) Describe the Cassegrain method of feeding a parabolic reflector with a neat diagram.
5(CO4)
3. (a) Draw and explain micro strip patch antenna with line feed. State its disadvantages.
5(CO4)
- (b) Explain general characteristics of Log periodic antenna with diagram.
5(CO4)
4. (a) What are different configurations of Antenna Array ? Explain any one of them.
5(CO5)

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- (b) Sketch the radiation pattern of a two element array having identical radiators spaced $\lambda/4$ apart and current in one radiator lags behind other by 90° . 5(CO5)
5. (a) Explain the features of various Smart Antenna systems. 7(CO2)
(b) What do you understand by Antenna Beamforming ? 3(CO2)
6. (a) Explain phenomenon of signal propagation in Smart Antenna System. 7(CO3)
(b) Elaborate in brief features of MIMO Antenna Systems. 3(CO3)

